Utkarsh Sinha

Data Scientist & Generative AI Engineer

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PROFESSIONAL SUMMARY

Data Scientist & Generative AI Engineer with 3+ years of experience shipping production AI systems across multi-agent LLM workflows, RAG, NLP, time-series forecasting and MLOps. Builds LangGraph agents on Databricks and GCP, fine-tunes open LLMs (32B–70B) with LoRA, and deploys RAG applications on AWS.

SKILLS

ML & Deep Learning: Python | PyTorch | CNN | RNN/LSTM | Transformers | BERT | LightGBM | Time-Series Forecasting
Generative AI: RAG | Fine-tuning (LoRA, Unsloth) | AI Agents | Embeddings | OpenAI | Claude | Gemini | LLaMA | Hugging Face
LLM Frameworks & APIs: LangChain | LangGraph | LlamaIndex | AWS Bedrock | LangSmith | Langfuse | FastAPI | Flask | Celery
Data & Vector Stores: PostgreSQL + pgvector | MySQL | MongoDB | Redis | FAISS | ChromaDB | Pinecone | LanceDB
MLOps & Cloud: Git | Docker | DVC | MLflow | GitHub Actions | Terraform | Prometheus | Grafana | AWS | Databricks | GCP
Vibe & Agentic Coding: Claude Code | Antigravity | Cursor

PROFESSIONAL EXPERIENCE

Data Scientist & GenAI Engineer *May 2026 – Present*
Aztra AI

- Build production multi-agent GenAI and ML systems for enterprise clients in retail demand planning and career-tech on Databricks and GCP, with MLflow tracing and CI/CD across develop, staging and production.
- Drive spec-driven agentic development with Claude Code and GitHub Spec Kit across 6 sprints with a test-first workflow.

ML & AI Engineer *December 2025 – March 2026*
SmartXAlgo

- Architected an AI stock-trading automation platform combining time-series forecasting, NLP sentiment and LLM reasoning, achieving ~70% directional accuracy in backtesting.
- Built a multi-model decision engine with ensemble approval and three additional AI-agent workflows, and automated CI/CD for reproducible model releases.

Data Scientist & GenAI Engineer *May 2023 – November 2025*
LawSikho & Skill Arbitrage

- Led a team of 2+ AI engineers building production ML and GenAI systems for education and HR tech.
- Designed MLOps pipelines with MLflow, DVC (AWS S3), Docker and GitHub Actions, cutting deployment time by ~40%; added Prometheus/Grafana monitoring that halved drift-detection latency.
- Delivered 1,000+ live training sessions on Data Science, MLOps and GenAI with a 4.5+/5 rating.

PROJECTS

Aurora V2 – Self-Improving Demand-Forecasting Agents *Aztra AI*

- Built 3 cooperating LangGraph agents (feature engineering, self-improving forecasting, model-health monitoring) for a fashion-retail demand-planning platform on Databricks with Claude Sonnet and MLflow 3.
- Designed a self-improving loop training cohort × horizon LightGBM models with Optuna and LLM-proposed configs, reaching 0.194 portfolio WAPE, plus counterfactual attribution of forecast misses.

Lyftrr – AI Career-Trajectory & Mentoring Platform *Aztra AI*

- Built a FastAPI + Celery AI service on Vertex AI Gemini generating career paths, roadmaps and adaptive skill assessments, shipped to GCP Cloud Run with Terraform.
- Engineered hybrid retrieval on PostgreSQL + pgvector fusing full-text, fuzzy, dense and SPLADE search via Reciprocal Rank Fusion, with an LLM generate-on-miss fallback.

Stock Market AutoTrader – Multi-Model AI Trading System *SmartXAlgo*

- Trained a Transformer multi-horizon forecaster on 5+ years of minute-level OHLCV data for 2,000+ equities, lifting directional consistency to ~80% with volatility, momentum and regime features.
- Fine-tuned BERT for financial news sentiment (~93% accuracy) and 32B Qwen/DeepSeek models with LoRA (Unsloth), cutting GPU memory by ~60%.

TalentLens – RAG-Powered CV Matching Engine *LawSikho & Skill Arbitrage*

- Matched 1,000+ CVs with hybrid embedding + TF-IDF scoring and LLaMA-70B reranking via Groq, cutting matching time by ~70% versus manual screening; deployed on AWS EC2 with Docker.

AWARDS

The Innovation Maverick – LawSikho & Skill Arbitrage *November 2023*

EDUCATION

BBA LLB – Guru Gobind Singh Indraprastha University Cumulative Percentage: 66.0%